

PROBLEM STATEMENT

Depth First Search (DFS) is a graph traversal technique used to explore all the vertices and edges of a graph. The objective is to visit every reachable vertex starting from a given source vertex by exploring one path completely before backtracking to explore other paths. DFS can be applied to both directed and undirected graphs and is widely used in applications such as path finding, cycle detection, maze solving, topological sorting, and connected component identification.